

District 6 Model Comparison - Geospatial Data: 2023 - 2025 Sales

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Model I

Table 1: Model Metrics: LN-SPPSF Model Comparison

Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	194	181	12	0.4577	0.2094	0.1570	0.5010	3.9961	296	342
+ Geospatial	194	182	11	0.5021	0.2521	0.2069	0.4859	5.5777	284	326
% Difference	0.00%	0.55%	-8.33%	9.72%	20.37%	31.77%	-3.00%	39.58%	-4.05%	-4.68%

Table 2: Ratio Metrics: LN-SPPSF Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9560	1.1262	0.8946	1.2589	44.5853	-0.7693	18.04%	33.51%	70.62%
+ Geospatial	0.9748	1.1188	0.8996	1.2437	42.0313	-0.6705	19.07%	37.11%	73.71%
% Difference	1.97%	-0.66%	0.56%	-1.21%	-5.73%	12.84%	5.71%	10.77%	4.38%

Model II

Table 3: Model Metrics: LN-PRICE (No ECF) Model Comparison

Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	299	283	15	0.6644	0.4415	0.4118	0.5483	14.9114	506	569
+ Geospatial	299	283	15	0.6767	0.4579	0.4292	0.5402	15.9356	497	560
% Difference	0.00%	0.00%	0.00%	1.84%	3.72%	4.20%	-1.48%	6.87%	-1.78%	-1.58%

Table 4: Ratio Metrics: LN-PRICE (No ECF) Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9935	1.1483	0.8712	1.3181	47.2291	-0.3816	15.38%	28.43%	62.88%
+ Geospatial	0.9828	1.1452	0.8756	1.3079	47.3274	-0.3637	14.38%	27.09%	65.22%
% Difference	-1.07%	-0.28%	0.51%	-0.78%	0.21%	4.69%	-6.52%	-4.71%	3.72%

Model III

Table 5: Model Metrics: LN-PRICE Model Comparison

Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	299	273	25	0.7369	0.5430	0.5011	0.5050	12.9745	466	566
+ Geospatial	299	274	24	0.7369	0.5430	0.5030	0.5040	13.5662	464	560
% Difference	0.00%	0.37%	-4.00%	0.00%	0.01%	0.37%	-0.19%	4.56%	-0.43%	-1.06%

Table 6: Ratio Metrics: LN-PRICE Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	1.0000	1.1204	0.9076	1.2345	41.7982	-0.2792	15.38%	32.44%	71.91%
+ Geospatial	1.0047	1.1207	0.9063	1.2366	41.2571	-0.2747	17.73%	31.44%	72.58%
% Difference	0.47%	0.03%	-0.14%	0.16%	-1.29%	1.62%	15.22%	-3.09%	0.93%

Model IV

Table 7: Model Metrics: LN-PRICE (Inliers) Model Comparison

Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	247	225	21	0.8015	0.6423	0.6090	0.3950	19.2430	265	345
+ Geospatial	249	226	22	0.8220	0.6758	0.6442	0.3853	21.4102	255	339
% Difference	0.81%	0.44%	4.76%	2.57%	5.20%	5.79%	-2.46%	11.26%	-3.77%	-1.74%

Table 8: Ratio Metrics: LN-PRICE (Inliers) Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9825	1.0736	0.9434	1.1380	32.7040	-0.1954	19.43%	39.27%	82.59%
+ Geospatial	1.0271	1.0692	0.9487	1.1269	30.0089	-0.1676	23.29%	42.17%	85.14%
% Difference	4.54%	-0.41%	0.56%	-0.97%	-8.24%	14.26%	19.86%	7.38%	3.09%

Model V

Table 9: Model Metrics: LN-TASP Model Comparison

Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	247	225	21	0.8001	0.6402	0.6066	0.3962	19.0636	266	347
+ Geospatial	249	226	22	0.8191	0.6709	0.6388	0.3882	20.9391	259	343
% Difference	0.81%	0.44%	4.76%	2.37%	4.79%	5.31%	-2.03%	9.84%	-2.63%	-1.15%

Table 10: Ratio Metrics: LN-TASP Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9989	1.0742	0.9454	1.1363	31.8707	-0.1996	23.08%	44.53%	80.57%
+ Geospatial	1.0002	1.0714	0.9448	1.1340	31.1270	-0.1718	23.69%	39.36%	83.94%
% Difference	0.13%	-0.27%	-0.06%	-0.20%	-2.33%	13.97%	2.68%	-11.62%	4.18%

Model VI

Table 11: Model Metrics: LN-TASP Model II Comparison

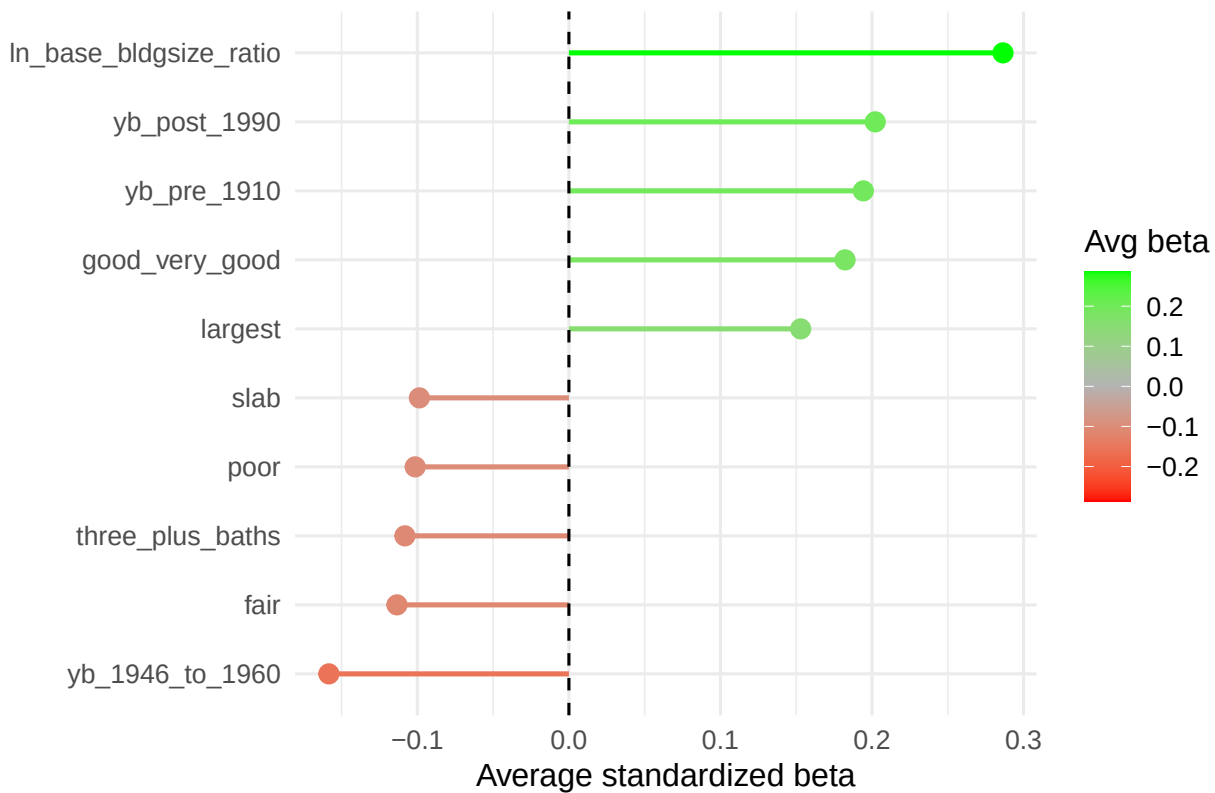
Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	247	226	20	0.7957	0.6332	0.6007	0.3991	19.5052	269	346
+ Geospatial	249	226	22	0.8191	0.6709	0.6388	0.3882	20.9391	259	343
% Difference	0.81%	0.00%	10.00%	2.93%	5.95%	6.34%	-2.75%	7.35%	-3.72%	-0.87%

Table 12: Ratio Metrics: LN-TASP Model II Comparison

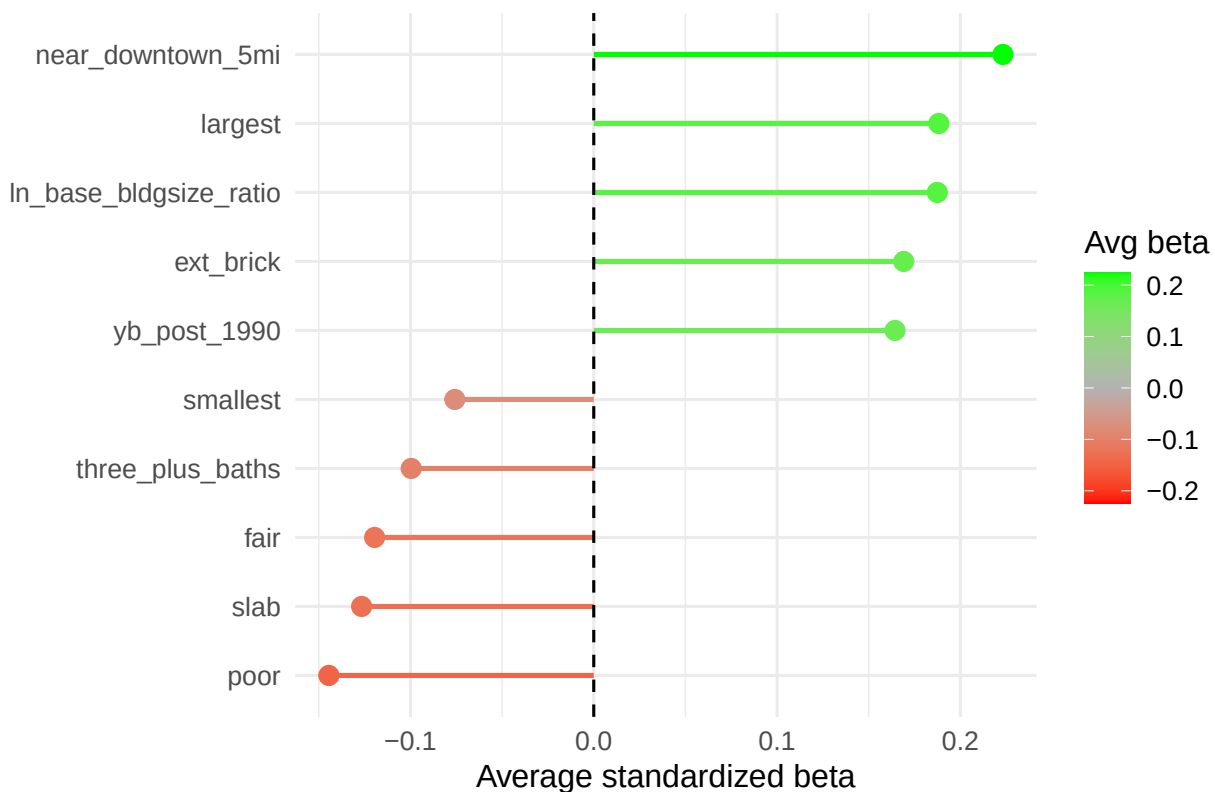
Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9874	1.0765	0.9421	1.1426	33.0138	-0.2022	21.46%	39.27%	81.38%
+ Geospatial	1.0002	1.0714	0.9448	1.1340	31.1270	-0.1718	23.69%	39.36%	83.94%
% Difference	1.29%	-0.48%	0.28%	-0.76%	-5.71%	15.03%	10.43%	0.22%	3.14%

Model Comparison

District 6 | Avg Std Beta across Initial Models



District 6 | Avg Std Beta across Geospatial Models



Prediction Testing

Table 13: Deviation Summary: Initial LN-SPPSF Model

Metric	Value
Average sale price deviation	-7936.41
Max deviation	166721.08
Min deviation	-545596.41
Standard deviation	67687.53

Table 14: Deviation Summary: LN-SPPSF Model + Geospatial

Metric	Value
Average sale price deviation	-7905.16
Max deviation	135785.33
Min deviation	-575766.50
Standard deviation	68234.40